

# Designed Student Turns for the Personas

## Strong student

Follows tutor's guidance well, gives mostly correct answers and occasional minor slips.

Question 1: Evaluate  $8 - 2 * 3 + 4$

1. How can I solve the following question  $8 - 2 * 3 + 4$ ?
2. I need to start with multiplication,  $2 * 3 = 6$
3. Then I subtract 6 from 8,  $8 - 6 = 2$
4. It remains  $2 + 4$ ,  $2 + 4 = 6$ , the answer is 6
5. I had to respect order of operations and do the multiplication before any addition or subtraction.
6. Yes — once the multiplication is done first, the expression evaluates cleanly to 6.

Question 2: Calculate  $3/4 - 1/6$

1. How do I solve  $3/4 - 1/6$ ?
2. The common denominator is 12
3. So I get  $3/4 = 3/4 * 3/3 = 9/12$
4.  $1/6 = 1/6 * 2/2 = 2/12$
5. Then subtraction,  $9/12 - 2/12 = 7/12$ , the final answer is  $7/12$
6. The key was converting both fractions to the same denominator so the numerators could be subtracted directly.
7. Right,  $7/12$  is the final result.

Question 3: Evaluate  $-3^2 + (-2)^3$

1. How do I solve  $-3^2 + (-2)^3$ ?
2.  $-3^2 = -9$
3.  $(-2)^3 = -8$
4.  $-9 + -8 = -17$ , the answer is -17
5. Confirmed, the final answer is -17
6. The important point is that the negative sign sits outside the square, so  $-3^2$  equals -9, not 9.

Question 4: Solve for x:  $2/3 * x = 10$

1. I need help with the following question:  $2/3 * x = 10$
2. I multiply the reciprocal of  $2/3$  on both sides of the equation:  $x = 10 * 3/2$
3.  $x = 30/2$ , therefore the answer is 15
4. Multiplying both sides by the reciprocal is what isolates x cleanly.
5. Yes — using the reciprocal of  $2/3$  gives  $x = 15$ .

Question 5: Solve for y:  $15 - 3y = 27$

1. Could you help me solve the following question:  $15 - 3y = 27$ ?
2. I subtract 15 on the both sides of the equation.
3.  $3y = 12$
4. It's not correct, actually  $-3y = 12$
5. Then I divide -3 on the both sides,  $y = -12/3$ , the final answer is -4
6. I needed to keep the negative sign with the y term the whole way through.
7. Checking:  $15 - 3(-4) = 15 + 12 = 27$ , so -4 is correct.

Question 6: Evaluate:  $1/2 + 3/5 / 9/10 - 1/3$

1. How can I simplify  $1/2 + 3/5 / 9/10 - 1/3$ ?
2. I should start with the division first, i.e.,  $3/5 / 9/10$
3.  $3/5 / 9/10 = 3/5 * 10/9 = 2/3$
4.  $1/2 + 2/3 = 3/6 + 4/6 = 7/6$
5.  $7/6 - 1/3 = 7/6 - 2/6 = 5/6$ , the answer is  $5/6$
6. Divisions and multiplications should be operated before additions and subtractions.
7. A common denominator is required for fraction additions and subtractions.

Question 7: Simplify:  $(1 + 1/2) / (1 - 3/4)$

1. How can I simplify  $(1 + 1/2) / (1 - 3/4)$ ?
2.  $1 + 1/2 = 2/2 + 1/2 = 3/2$
3. Then  $1 - 3/4 = 4/4 - 3/4 = 1/4$
4.  $3/2 / 1/4 = 3/2 * 4/1 = 6$ , the answer is 6
5. Parentheses first, then rewrite the division as multiplication by the reciprocal.
6. Yes, that gives 6.

Question 8: Solve for a:  $(a + 4)/3 - (a - 2)/2 = 1$

1. How do I solve  $(a + 4)/3 - (a - 2)/2 = 1$ ?
2. I start from finding a common denominator, the least common denominator is 6
3.  $(a + 4)/3 = (a + 4)/3 * 2/2 = 2(a + 4)/6$
4.  $(a - 2)/2 = (a - 2)/2 * 3/3 = 3(a - 2)/6$
5. The equation can be written as  $2(a + 4)/6 - 3(a - 2)/6 = 6/6$
6. So  $2(a + 4) - 3(a - 2) = 6$
7. I distribute the 2 and 3 into the parentheses,  $2a + 8 - 3a - 6 = 6$
8. It should be  $2a + 8 - 3a + 6 = 6$
9.  $-a + 14 = 6$
10. Then I subtract 14 on the both sides,  $-a = -8$ ,  $a = 8$ , the answer is 8
11. The critical slip was the sign when distributing  $-3(a - 2)$ ; it must become  $-3a + 6$ .
12. After fixing that sign and using the common denominator,  $a = 8$ .

Question 9: Solve for x:  $(2x - 1)/3 - (x + 3)/4 = (x + 1)/6 + 1/2$

1. How to solve the following question:  $(2x - 1)/3 - (x + 3)/4 = (x + 1)/6 + 1/2$ ?
2. I have to find the common denominator. 12 is the common denominator for 3, 4, 6 and 2
3. Then  $(2x - 1)/3 = (2x - 1)/3 * 4/4 = 4(2x - 1)/12$
4.  $(x + 3)/4 = (x + 3)/4 * 3/3 = 3(x + 3)/12$
5.  $(x + 1)/6 = (x + 1)/6 * 2/2 = 2(x + 1)/12$
6.  $1/2 = 1/2 * 6/6 = 6/12$
7. The equation will be  $4(2x - 1)/12 - 3(x + 3)/12 = 2(x + 1)/12 + 6/12$
8. Then I get  $4(2x - 1) - 3(x + 3) = 2(x + 1) + 6$
9. On the left side:  $8x - 4 - 3x - 9$ , which is  $5x - 13$
10. On the right side:  $2x + 2 + 6$ , which is  $2x + 8$
11. The equation can be written as  $5x - 13 = 2x + 8$
12. Add 13 on both sides to get  $5x = 2x + 21$
13. Subtract 2x on both sides:  $3x = 21$ ,  $x = 7$ , the answer is 7
14. Distributing the negative through the second set of parentheses was essential.
15. Once everything was over the common denominator 12, the equation simplified cleanly to  $x = 7$ .

Question 10: Solve for x:  $16^{(x-2)} = (1/8)^{(2x + 1)}$

1. I need help with solving  $16^{(x-2)} = (1/8)^{(2x + 1)}$
2.  $16 = 2^4$  and  $1/8 = 2^{-3}$
3. Then the equation can be written as  $2^4(x - 2) = 2^{-3}(2x + 1)$
4. Then I get  $4(x - 2) = -3(2x + 1)$
5. To distribute the factors in the parentheses, on the left side:  $4x - 8$
6. On the right side:  $-6x - 3$
7. The equation becomes  $4x - 8 = -6x - 3$
8. To add 6x and 8 on both side of the equation:  $10x = 5$
9. The answer is  $1/2$
10. Rewriting both sides with the same base (base 2) was the key step that turned the exponential equation into a linear one.
11. Yes,  $1/2$  is the solution.

## Average student

needs tutor's scaffolding, makes common errors and recovers with tutor's help.

Question 1: Evaluate  $8 - 2 * 3 + 4$

1. How can I solve the following question  $8 - 2 * 3 + 4$ ?
2. I start from the left,  $8 - 2 = 6$ , then multiply

3. Why it's not correct?
4. You mean to start from the multiplication? In that case,  $2 * 3 = 6$
5. So then from left to right, meaning  $8 - 6$  first
6.  $8 - 6 = 2$
7.  $2 + 4 = 6$ , the final answer is 6
8. I shouldn't start from left to right, I have to multiply first instead.
9. So I start from  $2 * 3$  in this case.

Question 2: Calculate  $3/4 - 1/6$

1. How do I solve  $3/4 - 1/6$ ?
2. So I need to calculate  $(3 - 1) / (4 - 6) = -1$
3. I'm not sure, is 16 their common denominator?
4. Would 12 work?
5.  $3/4 = 3/4 * 3/3 = 9/12$
6.  $1/6 = 1/6 * 2/2 = 2/12$
7. I only need to subtract the numerator, that is  $9/12 - 2/12 = (9 - 2)/12$ , the final answer is  $7/12$
8. The fractions needs to have common denominator in order to subtract the numerators.
9. Common denominator should be a number which can be divided by both denominators.

Question 3: Evaluate  $-3^2 + (-2)^3$

1. How do I solve  $-3^2 + (-2)^3$ ?
2. I'm not entirely sure,  $(-2)^3 = -8$ ?
3. Then  $-3^2 = 9$ ,  $9 - 8 = 1$
4. Negative? But the -3 is squared
5. I see, since the negative is outside the square,  $-3^2 = -9$
6. Then calculate  $-9 + -8$ , so the answer is -17
7. Yes, the final answer should be -17 instead of 1
8. I have to watch out for the negatives that are outside the square

Question 4: Solve for x:  $2/3 * x = 10$

1. I need help with the following question:  $2/3 * x = 10$
2. To cancel  $2/3$ , I multiply  $2/3$  on both sides of the equation:  $x = 10 * 2/3 = 20/3$
3. No? What is a reciprocal?
4. So the reciprocal of  $2/3$  is  $3/2$
5. If I multiply  $3/2$  on both sides of the equation, I'll get  $x = 10 * 3/2$
6.  $x = 15$ , the final answer is 15
7. To cancel a fraction factor, I have to multiply its reciprocal.
8. 15 is the right answer and  $20/3$  is incorrect

Question 5: Solve for y:  $15 - 3y = 27$

1. Could you help me solve the following question:  $15 - 3y = 27$ ?
2. I subtract 15 on the both sides of the equation, and I'll get  $15 - 3y + 15 = 27 + 15$
3. Let me check, you are right, I made a mistake,  $15 - 3y - 15 = 27 - 15$
4. Then it will be  $-3y = 12$
5. Then I divide -3 on the both sides,  $y = 12/3$ , the final answer is 4
6. Ah, the negative sign, the final answer is -4
7. I should subtract 15 from the both sides instead of addition
8. I have to pay attention to the negative signs during calculations.

Question 6: Evaluate:  $1/2 + 3/5 / 9/10 - 1/3$

1. How can I simplify  $1/2 + 3/5 / 9/10 - 1/3$ ?
2. I'll calculate from left to right,  $1/2 + 3/5 = 5/10 + 6/10 = 11/10$
3. No? I see, so instead I need to calculate  $3/5 / 9/10$  first. Could you give me a hint?
4. Okay, reciprocal,  $3/5 / 9/10 = 3/5 * 10/9 = 30/45$
5. Why I have to simplify before the next operation?
6.  $30/45 = 2/3$ , so it becomes  $1/2 + 2/3 - 1/3$
7. The common denominator for 2 and 3 is 6
8.  $1/2 + 2/3 - 1/3 = 3/6 + 4/6 - 2/6 = (3 + 4 - 2)/6$
9.  $3 + 4 - 2 = 5$ , hence the answer is  $5/6$
10. I have to do the division instead of calculating from left to right.

11.  $5/6$  is the right answer to the question.

Question 7: Simplify:  $(1 + 1/2) / (1 - 3/4)$

1. How can I simplify  $(1 + 1/2) / (1 - 3/4)$ ?
2. I can cancel 1s on the both sides, and get  $1/2 / 3/4$
3. 1s cannot be cancelled. I see. Then  $1 + 1/2 = 2/2 + 1/2 = 3/2$
4.  $1 - 3/4 = 4/4 - 3/4 = 1/4$
5. Then I get  $3/2 / 1/4$ , the common denominator is 4
6. Ah, I see. Not common denominator, but to find out the reciprocal
7. The reciprocal of  $1/4$  is  $4/1$
8.  $3/2 / 1/4 = 3/2 * 4/1 = 6$ , the answer is 6
9.  $2/3$  is not the correct answer to this question, 6 is the correct answer
10. The 1s inside the parentheses cannot be cancelled. The calculations instead the parentheses should start first.

Question 8: Solve for a:  $(a + 4)/3 - (a - 2)/2 = 1$

1. How do I solve  $(a + 4)/3 - (a - 2)/2 = 1$ ?
2. I start from finding a common denominator, the least common denominator is 6
3.  $(a + 4)/3 = (a + 4)/3 * 2/2 = 2(a + 4)/6$
4. Distributing 2 into the parentheses, I get  $(2a + 8)/6$
5. Similarly,  $(a - 2)/2 = (a - 2)/2 * 3/3 = 3(a - 2)/6 = (3a - 6)/6$
6. The equation can be written as  $(2a + 8)/6 - (3a - 6)/6 = 1$
7. As the common denominator is 6, the equation can be written as  $2a + 8 - 3a - 6 = 1$
8. 1 is not a fraction, why it has to be 6 instead?
9. I see, in that case,  $2a + 8 - 3a - 6 = 6$
10. You are right, I forgot the negative, so  $2a + 8 - 3a + 6 = 6$
11. I calculate a terms first,  $-a + 8 + 6 = 6$
12. Then I subtract 6 on the both sides:  $-a + 8 = 0$
13. Then I subtract 8 on the both sides,  $-a = -8$ , the answer is 8
14. Because  $1 = 6/6$ , that ensures the cancelation of the denominators on both sides of the equation
15.  $(8 + 4)/3 - (8 - 2)/2 = 12/3 - 4/2 = 1$ , 8 is the answer to the equation.

Question 9: Solve for x:  $(2x - 1)/3 - (x + 3)/4 = (x + 1)/6 + 1/2$

1. How to solve the following question:  $(2x - 1)/3 - (x + 3)/4 = (x + 1)/6 + 1/2$ ?
2. The first step, maybe finding the common denominator? I'm not entirely sure.
3. The common denominator, 24?
4. But 24 works, right, why I need to use a smaller number?
5. I see. Hmm, 12?
6. Good, then I get  $(2x - 1)/3 = (2x - 1)/3 * 4/4 = 4(2x - 1)/12$
7.  $(x + 3)/4 = (x + 3)/4 * 3/3 = 3(x + 3)/12$  and similarly  $(x + 1)/6 = (x + 1)/6 * 2/2 = 2(x + 1)/12$
8. The expression will be  $4(2x - 1)/12 - 3(x + 3)/12 = 2(x + 1)/12 + 1/2$
9.  $1/2$ ? But there's no variable with  $1/2$
10. Got it,  $4(2x - 1)/12 - 3(x + 3)/12 = 2(x + 1)/12 + 6/12$
11. So I get  $4(2x - 1) - 3(x + 3) = 2(x + 1) + 6$
12. Then I expand  $4(2x - 1)$ , which is  $8x - 4$
13. Similarly, expand  $3(x + 3)$ , which is  $3x + 9$
14. Negative? True,  $-3x - 9$
15. Then I expand also on the right side of the equation  $2x + 2 + 6$
16. The equation becomes  $8x - 4 - 3x - 9 = 2x + 2 + 6$
17. Calculate the numbers on the both sides to get  $8x - 3x - 13 = 2x + 8$ , then add 13 on both sides  $8x - 3x = 2x + 21$
18. Add  $3x$  and subtract  $2x$  on both sides to make the variable to be on the left side,  $3x = 21$
19.  $x = 21/3$ , so the answer is 7
20. I needed to find common denominator for the non-variable terms, otherwise the result is wrong
21. I have to watch out for the negative when distribute numbers inside parentheses.

Question 10: Solve for x:  $16^{(x-2)} = (1/8)^{(2x + 1)}$

1. I need help with solving  $16^{(x-2)} = (1/8)^{(2x + 1)}$

2. 16 can be written as  $4^2$ , and  $1/8$ , I don't know, I know that 8 can be written as  $4 * 2$
3. If I use 2 as base, then  $16 = 2^4$ . I don't know how to rewrite  $1/8$ , I know  $8 = 2^3$
4. Negative exponent rule? What should I do?
5. I see, you mean  $1/8 = 2^{(-3)}$ , right?
6. So the equation is now  $2^4(x - 2) = 2^{(-3)}(2x + 1)$ , what should I do next?
7. So I only keep the exponent, and I get  $4(x - 2) = -3(2x + 1)$
8. Then I expand,  $4x - 2 = -6x + 1$
9. Got it, then left side is  $4x - 8$
10. You are right, I need to multiply 3 and 1 too, so on the right side,  $-6x + 3$
11. True, I need to distribute the negative too,  $-6x - 3$
12. So the equation is now  $4x - 8 = -6x - 3$
13. And I move the variable to the left side by adding  $6x$  on the both sides,  $10x - 8 = -3$
14. Then move the numbers to the right side by adding 8,  $10x = 5$
15. And  $x = 10/5$ , the answer is 2
16. No, the answer is actually  $1/2$
17. Negative exponent can be used to rewrite fractions.
18. I forgot to distribute the factors to the second term in the parentheses, and that can cause wrong result.

## Struggling student

frequent misconceptions, sends confident-but-wrong answers to the tutor, triggers off-topic talk and abrupt topic changes.

Question 1: Evaluate  $8 - 2 * 3 + 4$

1. I need help with the following question  $8 - 2 * 3 + 4$ , the answer should be 22.
2. Of course the calculation starts from left to right  $8 - 2 * 3 + 4 = 6 * 3 + 4 = 22$
3. Then why it's wrong?
4. Start from multiplication? You mean  $2 * 3 = 6$ ?
5. Okay, then  $8 - 6 + 4$ , from left to right,  $8 - 6 = 2$
6. Then  $2 + 4 = 6$ , that's why they all say 6 is the correct answer.
7. I have to start with  $2 * 3$  first instead of calculating from  $8 - 2$
8. The reason why I thought 22 correct is I started from  $8 - 2$ , 6 is the correct answer.

Question 2: Calculate  $3/4 - 1/6$

1. How do I solve  $3/4 - 1/6$ ? Why the answer isn't -1?
2. Common denominator? What's common denominator?
3. Both 4 and 6 can be divided, like 2?
4.  $4 + 6 = 10$ , so that's the common denominator?
5. I see, then  $4 * 6 = 24$ , 24 is the common denominator.
6. So 12 it is, then  $3/4 = 3/4 * 1/3 = 3/12$
7. Not to change the value of the fraction, multiplying 1 won't change the value.  $1 = 3/3$
8. Then  $3/4 = 3/4 * 3/3 = 9/12$  and  $1/6 = 1/6 * 2/2 = 2/12$
9. Then  $9/12 - 2/12 = 7/10$
10. I see, denominator should be kept and only operate numerator. Then  $9/12 - 2/12 = (9 - 2)/12$  and the result is  $7/12$
11. Common denominator is a number that both denominators can be divided
12. The subtraction is only for the numerators

Question 3: Evaluate  $-3^2 + (-2)^3$

1. How do I solve  $-3^2 + (-2)^3$ ?
2. Math is my enemy. Anything you don't like to do?
3. Fine, left side. Just to be sure,  $^$  is not the same operator as  $*$ , correct?
4. Good, so  $-3^2 = -3 * (-3) = 9$ , and  $(-2)^3 = -2 * (-2) * (-2) = 8$
5. Pft, wrong? And you are telling me that the result of  $-3^2$  is a negative number?
6. So  $-3^2 = -(3) * (3) = -9$
7. The calculation of  $(-2)^3$  is wrong too?  $(-2)^3 = -2 * (-2) * (-2) = 4 * (-2) = -8$ , correct?
8. Then finally next step,  $-9 + -8$ ,  $9 + 8 = 17$  so the answer is -17
9. Go through the calculation again, fine,  $-3^2 = -9$  and  $(-2)^3 = -8$ , the sum of the two numbers is -17
10. The negative sign was the reason I couldn't get the correct answer.

Question 4: Solve for x:  $\frac{2}{3} * x = 10$

1. I need help with the following question:  $\frac{2}{3} * x = 10$
2. I'll minus 10 on the both side of the equation first,  $\frac{2}{3} * x - 10 = 0$
3. It was wrong? But I removed the same number from the both side of the equation.
4. What reciprocal? I don't get it. That sounds like a difficult word.
5. Flip the fraction? That sounds easy, and fun.
6.  $\frac{2}{3}$  flipped is  $\frac{3}{2}$ , and to multiply it on the both side of the equation.
7.  $\frac{2}{3} * x * \frac{3}{2} = 10 * \frac{3}{2}$ ,  $\frac{5}{5} * x = \frac{13}{2}$ , so the answer is  $\frac{13}{2}$
8. I can't add denominators and numerators, weird rules again.
9. Then  $\frac{6}{6} * x = \frac{30}{2}$ , then the answer is 15
10. So to go over the question again, I should multiply  $\frac{3}{2}$  on the both side of the equation, then I get  $x = \frac{30}{2} = 15$
11.  $\frac{3}{2}$  is reciprocal of  $\frac{2}{3}$

Question 5: Solve for y:  $15 - 3y = 27$

1. Could you help me solve the following question:  $15 - 3y = 27$ ?
2. I divide both side by 3, so  $15 - y = 9$ ,  $y = -6$
3. But I have to start with multiplication and division first instead of calculating from left to right.
4. I see. Then I need to cancel out 15 and 27 on the both side of the equation:  $-3y - 27 = -15$
5. Keep numbers and variable on different side. Then  $-3y = 27 - 15 = 12$
6. Then  $y = \frac{12}{3} = 4$ , the answer is 4
7.  $-3y = 12$ ,  $y = -\frac{12}{3} = -4$ , the answer is -4
8. I move 15 to the right side of the equation and keep  $-3y$  on the left side of the equation, that ensures the correct result.
9. I don't need to follow the left to right rule for solving this equation.

Question 6: Evaluate:  $\frac{1}{2} + \frac{3}{5} / \frac{9}{10} - \frac{1}{3}$

1. How can I simplify  $\frac{1}{2} + \frac{3}{5} / \frac{9}{10} - \frac{1}{3}$ ?
2. I start with  $\frac{1}{2} + \frac{3}{5} = \frac{4}{8}$ , then  $\frac{4}{8} / \frac{9}{10}$
3. Got it, we start from  $\frac{3}{5} / \frac{9}{10}$ ,  $\frac{3}{5} / \frac{9}{10} = (\frac{3}{9}) / (\frac{5}{10}) = \frac{1}{3} / \frac{1}{2}$
4. Do you know which star is the brightest in the night sky? Hint: it's not Polaris.
5. I don't know how to calculate fraction division.
6. Flipping the denominator and numerator to get the reciprocal? The reciprocal of  $\frac{3}{5}$  is  $\frac{5}{3}$ , and the reciprocal of  $\frac{9}{10}$  is  $\frac{10}{9}$ ,  $\frac{3}{5} / \frac{9}{10} = \frac{5}{3} * \frac{10}{9} = \frac{50}{27}$
7. Pft, wrong again. Okay, the second fraction only, so  $\frac{3}{5} / \frac{9}{10} = \frac{3}{5} * \frac{10}{9} = \frac{30}{45}$
8. Simplify? I don't get it. What should I do?
9. Common number that can divide 30 and 45,  $30 = 3 * 10$  and  $45 = 3 * 15$ , so  $\frac{30}{45} = \frac{10}{15}$
10. Not simplified enough. Okay, then  $10 = 5 * 2$ , and  $15 = 5 * 3$ , so  $\frac{10}{15} = \frac{2}{3}$
11. The expression is now  $\frac{1}{2} + \frac{2}{3} - \frac{1}{3}$ , and the answer is  $(1 + 2 - 1) / (2 + 3 - 3) = 1$
12. True, you are right, the calculation of fractions should start with finding a common denominator. So the common denominator is 6. Then  $\frac{1}{2} + \frac{2}{3} - \frac{1}{3} = \frac{3}{6} + \frac{4}{6} - \frac{2}{6}$
13. Now I add numerators only.  $3 + 4 - 2 = 5$
14. Then the denominator makes the final answer  $\frac{5}{6}$
15. I have to start with division first in this question instead of from left to right.
16. I cannot operate numerators and denominators separately in fraction calculations.

Question 7: Simplify:  $(1 + \frac{1}{2}) / (1 - \frac{3}{4})$

1. How can I simplify  $(1 + \frac{1}{2}) / (1 - \frac{3}{4})$ ?
2. I don't know, and I'm having headaches now.
3. Fine, the result is  $(\frac{1}{2}) / (\frac{3}{4})$ .
4. Because I simplified the fraction, so  $\frac{1}{2}$  remains as numerator and  $\frac{3}{4}$  remains as denominator.
5. But I should start with the division or multiplication.
6. OK, then parentheses first. Every time you tell me a new rule, annoying.
7.  $1 + \frac{1}{2} = \frac{3}{2}$ ,  $1 - \frac{3}{4} = \frac{1}{4}$
8. So the final answer is  $(\frac{3}{2}) / (\frac{1}{4})$
9. Not yet? But the numerator and denominator are clear now.
10. So now I have to simplify  $(\frac{3}{2}) / (\frac{1}{4})$ , what to do first?
11. Rewrite it to multiplication, so  $\frac{3}{2} * \frac{1}{4}$

12. Multiplication with the reciprocal,  $\frac{2}{3} * 4 = \frac{8}{3}$ , the answer is  $\frac{8}{3}$
13. Only flip the second fraction not both?  $\frac{3}{2} * 4$  instead?
14.  $\frac{3}{2} * 4 = \frac{12}{2}$ , the answer is  $\frac{12}{2}$
15. Simplify? Ah, okay.  $\frac{12}{2} = 6$ , the final answer is 6
16. I had to start with the calculations inside of the parentheses first.
17. I only needed to flip the second fraction when rewriting fraction division to multiplication.

Question 8: Solve for a:  $\frac{(a + 4)}{3} - \frac{(a - 2)}{2} = 1$

1. How do I solve  $\frac{(a + 4)}{3} - \frac{(a - 2)}{2} = 1$ ?
2. I move the numbers to the right side of the equation and keep the variable on the left side,  $(a + 4) - \frac{(a - 2)}{2} = 1 - \frac{1}{3} + \frac{1}{2}$
3. Not again! I followed the rules. Wait, so I need to start from the parentheses first? Then  $\frac{(a + 4)}{3} = \frac{a}{3} + \frac{4}{3}$ ,  $\frac{(a - 2)}{2} = \frac{a}{2} - \frac{2}{2}$ , so the equation becomes  $\frac{a}{3} + \frac{4}{3} - \frac{a}{2} - \frac{2}{2} = 1$
4. Then I can move the variables on the left side,  $\frac{a}{3} - \frac{a}{2} = 1 - \frac{4}{3} + \frac{2}{2} = -\frac{4}{3}$
5. Finding common denominator first? I want to have a cake now, or candy. Anything sweet.
6. Common denominator, what is it again? To flip the fraction?
7. So the number can be divided by 2 and 3, I see, 12 is the common denominator.
8. Why I have to find another one? 12 works, right?
9. Then 6. And the equation can be written as  $\frac{2(a + 4)}{6} - \frac{3(a - 2)}{6} = 1$
10. 1 needs to be written as a fraction with 6 as denominator too? I don't get it. 1 is a whole number.
11. You mean  $\frac{2(a + 4)}{6} - \frac{3(a - 2)}{6} = \frac{6}{6}$ , correct? Ah, so I can solve this equation instead,  $2(a + 4) - 3(a - 2) = 6$
12. Then I divide 3 and 2 on the both sides and then can move the numbers to the right side.
13. I should expand the terms inside parentheses instead? You mean to do something like  $2(a + 4) = 2a + 8$  and  $3(a - 2) = 3a - 6$
14. So the left side is  $2a + 8 - 3a - 6$ , the equation is  $2a + 8 - 3a - 6 = 6$ , then I can move the numbers to the right side and keep the variables on the left side.
15. Negative? That's annoying, that's why I hate math, there  $2a + 8 - 3a + 6 = 6$
16. Then I subtract 6 and 8 on the both sides  $2a - 3a = -8$
17. Left side of the equation is  $-a$ ,  $-a = -8$ , hence the answer is 8
18. Finding the common denominator should be the first step of this problem.
19. The whole number 1 should be considered as  $\frac{6}{6}$  in this question for simplifying expression.

Question 9: Solve for x:  $\frac{(2x - 1)}{3} - \frac{(x + 3)}{4} = \frac{(x + 1)}{6} + \frac{1}{2}$

1. How to solve the following question:  $\frac{(2x - 1)}{3} - \frac{(x + 3)}{4} = \frac{(x + 1)}{6} + \frac{1}{2}$ ?
2. I need to move  $\frac{(x + 1)}{6}$  to the left side of the equation,  $\frac{(2x - 1)}{3} - \frac{(x + 3)}{4} - \frac{(x + 1)}{6} = \frac{1}{2}$
3. Then I move the fractions to the right side,  $2x - x - x = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6}$ . Then I get  $0x$ , so  $x = 0$
4. Finding common denominator, huh, then the common denominator of 3 and 6 is 6, so I can rewrite  $\frac{(2x - 1)}{3}$  to  $\frac{2(2x - 1)}{6}$ .
5. I need to find a common denominator for 3, 4, 6 and 2 instead? But  $\frac{1}{2}$  is a fraction already, I don't need to find a common denominator for it.
6. I see. Then find a common denominator. 3, 2 and 6 have the common denominator 6, but 4 can't fit in.
7. Then the common denominator is 12, because  $4 + 6 = 10$ .
8. Yes, you're right, division not addition. Then the common denominator is 12. Then the equation is  $\frac{4(2x - 1)}{12} - \frac{3(x + 3)}{12} = \frac{2(x + 1)}{12} + \frac{6}{12}$ , then  $4(2x - 1) - 3(x + 3) = 2(x + 1) + 6$
9. Then distribute numbers inside the parentheses:  $8x - 1 - 3x + 9 = 2x + 2 + 6$ , move the numbers to one side and variables to another:  $8x - 3x - 2x = 2 + 6 + 1 - 9$
10. The distribution is wrong? You must be joking. I did it correctly this time. Check it again.
11. Negative was not distributed. Fine. Then  $8x - 1 - 3x - 9 = 2x + 2 + 6$ , correct?
12. 4 is not distributed correctly?  $4(2x - 1) = 8x - 4$ , I see. Then  $8x - 4 - 3x - 9 = 2x + 2 + 6$
13. Then continue with the moving the numbers and variables to the two sides,  $8x - 3x - 2x = 2 + 6 + 4 + 9$ , I'll quit if you keep adding additional calculation rules.
14. Then on the left side:  $8x - 3x - 2x = 3x$  and right side:  $2 + 6 + 4 + 9 = 18$ ,  $3x = 18$ , the answer is 6
15. Wrong calculation on right side? You kidding?  $2 + 6 = 8$ , and  $4 + 9 = 13$ ,  $8 + 13 = 21$ . Okay right side is 21
16. Then I get  $3x = 21$ ,  $x = 7$ , the answer is 7

17. The first step is to find the common denominator for the all the fractions in the equation.
18.  $(2 * 7 - 1)/3 - (7 + 3)/4 = 13/3 - 10/4 = 52/12 - 30/12 = 22/12$ ,  $(7 + 1)/6 + 1/2 = 8/6 + 1/2 = 16/12 + 6/12 = 22/12$ , 7 is the correct answer to the equation.

Question 10: Solve for x:  $16^{(x-2)} = (1/8)^{(2x+1)}$

1. I need help with solving  $16^{(x-2)} = (1/8)^{(2x+1)}$ . I multiply 1/16 on both sides of the equation to cancel 16 first then get  $x - 2 = (1/8)^{(2x+1)} * 1/16$ , right?
2. Wait, it can solve the problem, see, then I multiply 8 on both sides to cancel 1/8,  $8(x - 2) = (2x + 1) * 1/16 * 8$ ,  $8x - 16 = 1/2(2x + 1)$ ,  $7x = 16 + 1/2$ , so the  $x = 33/14$
3. I followed the rules, I multiplied same numbers on the both sides of the equation. And it's not right?
4. To rewrite the 16 and 1/8 to the same number?  $16 = 8 * (16)/8$ . 16 and 1/8 are not the same.
5. The base number for exponent. For instance?
6.  $4 = 2^2$ , I see. Then  $16 = 2^4$ , and  $1/8 = 1/(2^3)$ .
7. That's frustrating. I don't want to continue. Just give me the final answer and I can finish my assignments.
8.  $16 = 2 * 8 = 2 * 2 * 4 = 2 * 2 * 2 * 2 = 2^4$ , and  $8 = 2 * 2 * 2 = 2^3$
9. So  $2^4(x-2) = (1/(2^3))^{(2x+1)}$ , what should I do next?
10. Negative exponent,  $1/8 = 8^{(-1)} = 2^{(-3)}$ , then the equation is  $2^4(x-2) = 2^{(-3)(2x+1)}$
11. Then I can finally cancel 2s on the both sides of the equation. So I get  $4(x - 2) = -3(2x + 1)$
12. Then  $4x - 8 = -6 + 3$ , move the numerics to the right side  $4x = -6 + 3 + 8$
13. Mistake on distribution?  $4(x - 2) = 4x - 8$ , correct?
14.  $-3(2x + 1) = -6x - 3$ , so  $4x - 8 = -6x - 3$
15. Now finally solve the equation:  $4x + 6x = -3 + 8$
16. Then I get  $10x = -5$ , the answer is  $-1/2$
17. Sure, another mistake. It's exhausting to solve this problem.
18. Let's see,  $4x + 6x = 10x$ , and  $-3 + 8 = 5$ , so I get  $x = 5/10 = 1/2$ . The final answer is  $1/2$
19. I cannot simply cancel the numbers with exponents.
20.  $2^8$  is different from  $2*8$ . Because  $2^8$  means the multiplication of 2 for 8 times.